HUYNH THANH BAO NGOC

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LinkedIn Github Portfolio

SUMMARY

Computer Science background with knowledge of artificial intelligence, machine learning, data analysis, and data management. Experienced in academic and personal projects involving AI applications, predictive modeling, data visualization, and data-driven solutions. Seeking internship or entry-level opportunities in AI/ML and Data to apply technical and analytical skills while gaining practical experience.

EDUCATION

TDTU – Ton Duc Thang University **2022–2026**
Bachelor of Computer Science | GPA: 7.97/10

WORK EXPERIENCE

Full-stack Developer Intern – Department of Science and Technology (Vinh Long) **Feb – Apr 2026**

- Researched and developed a RAG-based AI chatbot to support administrative document processing for government officers.
- Built user and admin interfaces, enabling users to chat with the chatbot and administrators to upload and manage documents.

PROJECTS

Northwind Sales Analytics (AI Agent + BI + Chatbot) **Mar 2026**
Personal Project

- **Description:** Developed a sales analytics project using the Northwind dataset, integrating an AI chatbot to support natural language querying and business insight retrieval from sales data.
- **Responsibilities:**
  - Performed exploratory data analysis (EDA) to identify patterns in customer behavior and product performance.
  - Applied machine learning algorithms (e.g., K-Means clustering) to segment customers into meaningful groups.
  - Built Power BI dashboards to visualize sales, product, and customer insights.
  - Developed an n8n-based AI agent chatbot that translates natural language into SQL queries for data analysis and insight retrieval.
- **Technologies:** Python, Postgres, Pandas, Scikit-learn, Power BI, Machine Learning, n8n
- **Sources:** Git

Administrative Document Retrieval Chatbot (RAG-based) **Feb – Mar 2026**
Personal Project

- **Description:** Developed a RAG-based chatbot for administrative document search, allowing users to ask questions in natural language and receive accurate answers, with an admin interface to manage users and documents.
- **Responsibilities:**
  - Built frontend with React and backend with FastAPI, implementing a RAG pipeline for question answering.
  - Performed hybrid search on Milvus with re-ranking, followed by response generation using a local LLM.
  - Processed documents using OCR and segmentation, then generated vector embeddings for efficient retrieval.
  - Evaluated system performance through query scenarios and API testing, optimizing chunking strategies, retrieval parameters, re-ranking, and model performance.
- **Technologies:** Python, MongoDB, FastAPI, React, Milvus, Docker, Langchain, RAG, Docling, Ollama
- **Sources:** Git

Vietnamese News Classification **June 2026**
Personal Project

- **Description:** Developed a Vietnamese news classification system to categorize VnExpress articles and detect out-of-domain content.
- **Responsibilities:**
  - Preprocessed Vietnamese text by performing word segmentation, text normalization, and stopword removal.
  - Implemented and compared traditional machine learning (TF-IDF + SVM) and deep learning (PhoBERT) models for Vietnamese text classification.
  - Evaluated model performance using Accuracy and Macro-F1, with the best model achieving **92.59% Accuracy** and **92.88% Macro-F1**.
- **Technologies:** Python, Pandas, Scikit-learn, PyTorch, PhoBERT, Underthesea, NLP
- **Sources:** Git

SKILLS

Programming Languages: Python, Java, SQL
Databases: SQL Server, MySQL, PostgreSQL, MongoDB
Vector Databases: Milvus
DevOps & Tools: Docker, Github, Postman, PowerBI, Excel, n8n
AI/ML: PyTorch, TensorFlow, scikit-learn, LLM, RAG
Others: Agile/Scrum development
Soft Skills: Problem-solving, Teamwork, Communication, Time Management
Languages: Vietnamese (Native), English (Toeic 700)

CERTIFICATES

SQL (Intermediate) Certificate – HackerRank
MongoDB Basics for Students
Google Advanced Data Analytics Capstone